

DETAILED MEDICAL ENCYCLOPEDIA

A structured medical reference for anatomy, physiology, pathology, diseases, diagnostics, pharmacology and emergency medicine

Educational reference only — not a substitute for a licensed medical professional.

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1. Foundations of Medicine

Homeostasis and disease

Homeostasis is the maintenance of a relatively stable internal environment. Disease is a structural, functional, or biochemical abnormality that disrupts normal physiology. Clinical reasoning links symptoms and signs to anatomy, pathophysiology, differential diagnosis, investigations, and management.

Core clinical vocabulary

Anatomy describes structure; physiology describes normal function; pathology describes disease-related structural changes; pathophysiology describes altered function; pharmacology studies drugs; microbiology studies microorganisms; epidemiology studies disease distribution and determinants.

Clinical assessment

History commonly includes chief complaint, history of present illness, past medical history, medications, allergies, family history, social history, and review of systems. Examination may include general appearance, vital signs, cardiovascular, respiratory, abdominal, neurological and musculoskeletal assessment.

2. Human Anatomy

Anatomical organization

The body is organized from cells to tissues, organs, organ systems, and the whole organism. Major tissue classes are epithelial, connective, muscle, and nervous tissue.

Body systems

Major systems include nervous, cardiovascular, respiratory, digestive, urinary, reproductive, endocrine, immune/lymphatic, musculoskeletal, integumentary, and sensory systems.

Directional terms

Superior/inferior indicate toward the head/feet; anterior/posterior toward the front/back; medial/lateral toward or away from the midline; proximal/distal describe relative position along a limb; superficial/deep describe distance from the surface.

3. Physiology

Cardiovascular physiology

The heart generates pressure to circulate blood. Cardiac output equals heart rate $\times$ stroke volume. Blood pressure is influenced by cardiac output and systemic vascular resistance. Oxygen delivery depends on cardiac output and arterial oxygen content.

Respiratory physiology

Ventilation moves air; diffusion exchanges gases across the alveolar-capillary membrane; perfusion delivers blood to the lungs. Gas exchange is strongly influenced by ventilation-perfusion matching.

Renal physiology

The kidneys filter plasma, reabsorb needed substances, secrete selected compounds, regulate electrolytes and acid-base balance, and produce hormones including erythropoietin and renin.

Endocrine physiology

Hormones act through receptors to regulate metabolism, growth, reproduction, stress responses, fluid balance and other processes. Many endocrine systems use negative feedback.

4. Pathology & Immunology

Inflammation

Acute inflammation is characterized by rapid vascular and cellular responses, commonly involving neutrophils. Chronic inflammation may involve macrophages, lymphocytes, tissue destruction and repair/fibrosis.

Cell injury

Cell injury can result from hypoxia, toxins, infection, immune reactions, genetic abnormalities or physical injury. Reversible injury may progress to cell death if the insult persists.

Immunity

Innate immunity provides rapid, nonspecific defense. Adaptive immunity is antigen-specific and includes B cells, antibodies, T cells and immunological memory.

Neoplasia

Neoplasia is abnormal clonal cell proliferation. Benign tumors generally remain localized, whereas malignant tumors can invade and metastasize.

5. Infectious Diseases

Bacterial infection

Bacteria may cause localized or systemic disease through invasion, toxins, inflammation or immune-mediated damage. Diagnosis can involve microscopy, culture, antigen tests or molecular methods.

Viral infection

Viruses replicate inside host cells. Clinical manifestations range from asymptomatic infection to severe systemic disease. Antiviral treatment is pathogen-specific.

Fungal and parasitic disease

Fungal infections range from superficial disease to invasive infections, particularly in vulnerable hosts. Parasites include protozoa and helminths with diverse transmission cycles.

Antimicrobial principles

Antimicrobials should be selected according to the suspected/confirmed organism, site of infection, resistance patterns, patient factors and stewardship principles. Do not self-prescribe antibiotics.

6. Cardiovascular Medicine

Hypertension

Hypertension is persistently elevated blood pressure and is a major risk factor for stroke, coronary disease, heart failure and kidney disease. Diagnosis generally requires appropriate repeated measurements rather than a single

reading.

Coronary artery disease

Atherosclerotic narrowing of coronary arteries can cause stable angina or acute coronary syndromes. Myocardial infarction occurs when prolonged ischemia causes myocardial injury and necrosis.

Heart failure

Heart failure is a clinical syndrome in which the heart cannot meet physiological demands without elevated filling pressures. It may involve reduced or preserved ejection fraction.

Arrhythmias

Arrhythmias are abnormalities of cardiac rate, rhythm or conduction. Symptoms can include palpitations, dizziness, syncope or chest discomfort; some are asymptomatic.

7. Respiratory Medicine

Asthma

Asthma involves variable respiratory symptoms and airflow limitation associated with airway inflammation and hyperresponsiveness. Triggers may include allergens, infections, exercise and irritants.

COPD

Chronic obstructive pulmonary disease causes persistent airflow limitation, commonly associated with tobacco smoke or other inhalational exposures. Management includes risk-factor reduction, inhaled therapies, rehabilitation and selected oxygen therapy.

Pneumonia

Pneumonia is infection of the lung parenchyma. Presentation can include fever, cough, sputum production, dyspnea and abnormal chest findings. Severity assessment guides treatment and disposition.

Pulmonary embolism

Pulmonary embolism occurs when a thrombus, usually from the venous circulation, obstructs pulmonary arteries. Sudden dyspnea, pleuritic chest pain, tachycardia or syncope can occur and require urgent assessment.

8. Gastroenterology

GERD

Gastroesophageal reflux disease occurs when reflux of gastric contents causes troublesome symptoms or complications. Typical symptoms include heartburn and regurgitation.

Peptic ulcer disease

Peptic ulcers are mucosal defects extending into deeper tissue, commonly associated with *Helicobacter pylori* infection or NSAID exposure.

Liver disease

Liver disorders include hepatitis, fatty liver disease, cirrhosis and cholestatic disease. Chronic injury can lead to fibrosis, portal hypertension, liver failure and hepatocellular carcinoma.

Acute abdomen

Severe abdominal pain may result from appendicitis, obstruction, perforation, pancreatitis, biliary disease, ischemia or other causes. Peritonism, shock or severe progressive pain requires urgent evaluation.

9. Neurology

Stroke

Stroke is acute neurological dysfunction due to vascular injury. Ischemic stroke results from arterial occlusion; hemorrhagic stroke results from bleeding. Rapid recognition and emergency evaluation are critical.

Seizures and epilepsy

A seizure is transient abnormal neuronal activity; epilepsy is a disorder characterized by a predisposition to recurrent unprovoked seizures. Evaluation depends on the event characteristics and underlying cause.

Headache

Primary headaches include migraine and tension-type headache. Secondary headaches arise from another disorder. Sudden maximal-at-onset headache, new neurological deficit, altered consciousness or systemic red flags warrants urgent assessment.

Peripheral neuropathy

Peripheral neuropathy can produce numbness, tingling, burning pain, weakness or loss of reflexes. Diabetes, nutritional deficiencies, toxins, medications and immune disorders are among possible causes.

10. Endocrinology & Metabolism

Diabetes mellitus

Diabetes is characterized by chronic hyperglycemia due to impaired insulin secretion, insulin action, or both. Long-term complications include cardiovascular disease, kidney disease, neuropathy and retinopathy.

Thyroid disorders

Hypothyroidism commonly causes fatigue, cold intolerance, weight gain and constipation; hyperthyroidism can cause heat intolerance, weight loss, tremor and palpitations. Diagnosis uses thyroid function testing.

Adrenal disorders

Adrenal disease includes cortisol excess, cortisol deficiency and disorders of aldosterone or catecholamine production. Severe adrenal insufficiency can be life-threatening.

Electrolytes

Sodium, potassium, calcium, magnesium and phosphate abnormalities can affect neurological, muscular and cardiac function. Severe or rapidly changing abnormalities may be emergencies.

11. Nephrology & Urology

Acute kidney injury

Acute kidney injury is an abrupt decline in kidney function, identified using changes in serum creatinine and/or urine output. Causes are broadly categorized as pre-renal, intrinsic renal and post-renal.

Chronic kidney disease

CKD is persistent abnormal kidney structure or function. Common causes include diabetes and hypertension. Disease progression increases cardiovascular and renal risks.

Urinary tract infection

UTIs can involve the bladder or upper urinary tract. Dysuria, frequency and urgency suggest lower UTI; fever, flank pain or systemic illness may indicate pyelonephritis.

Kidney stones

Renal calculi may cause severe colicky flank pain, hematuria, nausea and vomiting. Fever with obstruction is a urological emergency.

12. Hematology & Oncology

Anemia

Anemia is reduced hemoglobin concentration relative to the relevant reference range. Evaluation considers cell size, reticulocyte response, bleeding, iron status, vitamin deficiency, chronic disease and hemolysis.

Clotting disorders

Hemostasis depends on platelets, coagulation factors, anticoagulant systems and fibrinolysis. Disorders may present with bleeding or thrombosis.

Leukemia and lymphoma

Leukemias are malignant disorders of blood-forming tissues; lymphomas arise primarily in lymphoid tissues. Diagnosis commonly requires blood, marrow, tissue and immunophenotypic studies.

Cancer principles

Cancer treatment may include surgery, radiation, chemotherapy, targeted therapy, immunotherapy, endocrine therapy or combinations. Treatment depends on tumor type, stage, molecular features and patient factors.

13. Musculoskeletal & Rheumatology

Osteoarthritis

Osteoarthritis is a degenerative joint disorder characterized by cartilage loss and structural changes. Pain typically worsens with activity and improves with rest.

Rheumatoid arthritis

Rheumatoid arthritis is an autoimmune inflammatory arthritis that can cause persistent symmetric joint inflammation and systemic manifestations.

Gout

Gout results from monosodium urate crystal deposition, producing acute inflammatory arthritis. Commonly affected joints include the first metatarsophalangeal joint.

Fractures

Fracture management depends on location, displacement, stability, soft-tissue injury, neurovascular status and patient factors. Open fractures and neurovascular compromise require urgent care.

14. Dermatology

Eczema/dermatitis

Dermatitis describes inflammatory skin disease characterized by itching, erythema, scaling or vesicles depending on type and stage.

Acne

Acne involves follicular plugging, sebum production, inflammation and microbial factors. Treatment is selected according to severity and lesion type.

Skin infection

Bacterial, fungal and viral infections can affect skin and soft tissue. Rapidly spreading redness, severe pain, systemic toxicity or tissue necrosis requires urgent evaluation.

Skin cancer

Common skin cancers include basal cell carcinoma, squamous cell carcinoma and melanoma. New, changing, bleeding or suspicious pigmented lesions should be clinically assessed.

15. Obstetrics & Gynecology

Pregnancy physiology

Pregnancy produces cardiovascular, respiratory, renal, hematologic and endocrine adaptations. Some laboratory and vital-sign changes are physiological, while others signal complications.

Hypertensive disorders of pregnancy

Gestational hypertension and preeclampsia are important pregnancy complications. Severe hypertension, neurological symptoms, visual symptoms, severe abdominal pain or seizures require urgent assessment.

Menstrual disorders

Abnormal uterine bleeding may result from structural lesions, ovulatory dysfunction, endocrine disorders, medications, pregnancy-related causes or malignancy.

Reproductive health

Contraceptive options include barrier, hormonal, intrauterine and permanent methods. Choice depends on effectiveness, contraindications, preferences and reproductive goals.

16. Pediatrics

Growth and development

Pediatric assessment includes growth parameters, developmental milestones, nutrition, immunization status and age-specific examination.

Common pediatric illness

Respiratory infections, gastroenteritis, fever and common rashes are frequent presentations. Age, hydration, respiratory effort and general appearance are important severity indicators.

Dehydration

Children can deteriorate quickly with fluid loss. Signs may include reduced urine output, dry mucosa, lethargy, tachycardia and poor perfusion depending on severity.

Pediatric emergencies

Breathing difficulty, cyanosis, seizures, altered consciousness, severe dehydration, shock and suspected serious infection require urgent medical evaluation.

17. Psychiatry

Depression

Depressive disorders involve persistent changes in mood, interest and functioning, with associated cognitive, sleep, appetite or energy symptoms. Diagnosis requires clinical assessment.

Anxiety disorders

Anxiety disorders involve excessive fear or worry and may produce physical symptoms such as palpitations, sweating, gastrointestinal symptoms and sleep disturbance.

Psychosis

Psychosis involves impaired reality testing, potentially including hallucinations, delusions and disorganized thought or behavior. New-onset psychosis requires medical and psychiatric assessment.

Safety

Suicidal thoughts, plans, intent, severe agitation, inability to care for oneself or dangerous behavior require immediate professional support and emergency assessment.

18. Emergency Medicine & First Aid

ABCDE approach

In an acutely ill patient, assess Airway, Breathing, Circulation, Disability and Exposure. Treat immediately life-threatening problems while continuing assessment.

CPR

For an unresponsive person who is not breathing normally, activate emergency medical services and begin CPR. Use an automated external defibrillator when available and follow its prompts.

Shock

Shock is inadequate tissue perfusion. Major categories include hypovolemic, distributive, cardiogenic and obstructive shock. Signs may include altered mental status, cool extremities, tachycardia and hypotension, although early shock may occur without hypotension.

Emergency red flags

Chest pain with concerning features, severe breathing difficulty, stroke symptoms, major bleeding, seizures, anaphylaxis, severe burns, poisoning, loss of consciousness and sudden severe neurological symptoms warrant emergency care.

19. Diagnostics & Laboratory Medicine

CBC

A complete blood count evaluates red cells, hemoglobin, white cells and platelets. Interpretation considers patterns rather than isolated values.

Basic metabolic testing

Electrolytes, glucose, renal markers and related tests help assess metabolic and organ function. Abnormal results must be interpreted in clinical context.

Imaging

X-ray is useful for many bone and chest conditions; ultrasound evaluates fluid, abdominal organs, vessels and pregnancy; CT provides rapid cross-sectional imaging; MRI offers high soft-tissue contrast.

ECG

Electrocardiography records cardiac electrical activity. Clinicians assess rate, rhythm, axis, intervals and ST-T changes in clinical context.

20. Pharmacology Principles

Pharmacokinetics

Pharmacokinetics describes what the body does to a drug: absorption, distribution, metabolism and excretion (ADME).

Pharmacodynamics

Pharmacodynamics describes what a drug does to the body, including receptor interactions, dose-response relationships, therapeutic effects and adverse effects.

Drug safety

Medication safety requires correct drug, dose, route, timing, patient and indication. Drug interactions, allergies, renal/hepatic function, pregnancy and age can alter risk.

Important warning

This encyclopedia is educational, not a prescribing guide. Drug names and doses should not be used for self-treatment. Prescription decisions require a qualified clinician and current authoritative references.

21. Preventive Medicine

Vaccination

Vaccines train adaptive immunity to reduce the risk of specific infections and complications. Schedules vary by age, risk, country and health status.

Screening

Screening tests asymptomatic populations to detect disease or risk early. Appropriate screening depends on age, risk factors, test performance and guideline recommendations.

Lifestyle

Evidence-based prevention includes avoiding tobacco, regular physical activity, balanced nutrition, adequate sleep, appropriate weight management, safer alcohol use and routine preventive care.

Public health

Public health addresses disease prevention at population level through surveillance, sanitation, vaccination, health education, environmental interventions and policy.

22. Clinical Reasoning Framework

Differential diagnosis

A differential diagnosis is a ranked list of plausible causes. Build it from anatomy, time course, epidemiology, risk factors, symptom patterns and examination findings.

Red flags first

Before pursuing a common benign diagnosis, actively exclude time-critical conditions. The cost of missing a dangerous diagnosis is often much higher than the cost of evaluating it.

Testing strategy

Tests should answer a clinical question. Consider pre-test probability, sensitivity, specificity, likelihood ratios, false positives, false negatives, harms and whether the result will change management.

Evidence and uncertainty

Medical knowledge changes. When making real clinical decisions, use current guidelines, drug labels, local protocols and qualified medical supervision rather than relying on a static encyclopedia.

Quick Reference: Seek Emergency Care

Seek emergency medical attention for symptoms such as severe or worsening chest pain, severe difficulty breathing, signs of stroke (facial droop, arm weakness, speech difficulty), uncontrolled major bleeding, loss of consciousness, prolonged or repeated seizures, severe allergic reaction with breathing difficulty, poisoning, severe burns, or sudden severe neurological deterioration.

This list is not exhaustive. If a person appears seriously ill or deteriorating rapidly, contact local emergency medical services.